Trust-Aware Review Intelligence for Reliable Service Provider Ranking Using Multi-Task Aspect-Based Sentiment Analysis

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***Abstract*—Online service marketplaces increasingly rely on customer reviews and numerical ratings to evaluate service providers and support customer decision-making. However, conventional rating-based approaches provide only an overall representation of service quality and may be influenced by suspicious, manipulated, or unreliable reviews. Furthermore, a single numerical rating cannot effectively represent customer opinions regarding different dimensions of service quality. To address these limitations, this paper proposes a Trust-Aware Review Intelligence Framework that integrates Multi-Task Aspect-Based Sentiment Analysis with Review Credibility Detection for reliable service provider evaluation and ranking.**

The proposed framework receives the Top-10 candidate service providers from the preceding recommendation stage and analyses their customer reviews across four service-quality aspects: Quality, Punctuality, Communication, and Professionalism. A Multi-Task Bidirectional Long Short-Term Memory (Bi-LSTM) network with Multi-Head Attention simultaneously predicts aspect-level sentiment by learning shared contextual representations across the four related sentiment classification tasks. In parallel, an Isolation Forest-based credibility detection module analyses linguistic and behavioural review characteristics to identify anomalous review patterns and estimate review credibility. The predicted aspect sentiment scores are combined through weighted aspect fusion to obtain an overall sentiment score, which is subsequently adjusted using the corresponding review credibility score through a Trust-Aware Sentiment Fusion mechanism. Review-level trust-aware scores are then aggregated for each candidate provider, enabling the framework to rank the Top-10 candidates and select the final Top-5 trustworthy service providers.

The framework was evaluated using 25,000 customer reviews distributed across fourteen service categories. Experimental results achieved testing accuracies of 96.10%, 87.33%, 85.48%, and 85.48% for Quality, Punctuality, Communication, and Professionalism, respectively, resulting in an average testing accuracy of 88.60%. These findings demonstrate the effectiveness of the proposed multi-task sentiment model in capturing aspect-specific customer opinions, while the integration of review credibility provides a trust-aware mechanism for reducing the influence of suspicious feedback during service provider evaluation.

Keywords- Aspect-Based Sentiment Analysis, Multi-Task Learning, Bidirectional Long Short-Term Memory (Bi-LSTM), Multi-Head Attention, Isolation Forest, Review Credibility Detection, Trust-Aware Sentiment Fusion, Service Provider Recommendation.

I. INTRODUCTION

Online service marketplaces have become increasingly popular for connecting customers with skilled service providers across various domains such as plumbing, electrical maintenance, carpentry, air conditioning repair, painting, vehicle repair, and household maintenance services. These platforms generate a large volume of customer reviews that significantly influence provider reputation and customer decision-making. Consequently, customer reviews have become one of the most valuable sources for evaluating service quality and recommending reliable service providers [3], [6].

Most existing recommendation systems primarily rely on provider profiles, numerical ratings, historical performance, and user interactions to generate provider recommendations

[3]. Although these approaches effectively identify potential service providers, they often overlook the rich information embedded within customer reviews. Reviews contain detailed customer experiences regarding different aspects of service quality that cannot be accurately represented using overall ratings alone [6], [7]. Furthermore, review manipulation, fake reviews, and opinion spam can distort provider rankings, reducing the reliability of recommendation results [8], [9], [13].

Within the proposed ML-Driven Service Provider Matching Platform, the Dynamic Credibility Verification module (Component 2) produces the Top-10 verified service providers based on provider credibility indicators. However, providers with similar credibility scores may still exhibit considerable differences in service quality when customer reviews are analyzed in detail. Therefore, an additional review intelligence mechanism is required to evaluate provider performance using aspect-level customer opinions while minimizing the influence of suspicious reviews.

To address this challenge, this paper proposes a Trust-Aware Review Intelligence Framework that functions as the final intelligence layer of the recommendation pipeline. The framework performs Multi-Task Aspect-Based Sentiment Analysis using a Bidirectional Long Short-Term Memory (Bi-LSTM) network enhanced with a Multi-Head Attention mechanism to simultaneously predict customer sentiment across four critical service quality aspects: *Quality, Punctuality, Communication,* and *Professionalism* [2], [4], [10], [11]. In parallel, an Isolation Forest-based anomaly detection model estimates the credibility of each customer review by identifying abnormal linguistic and behavioural patterns [8], [13]. Finally, a Trust-Aware Sentiment Fusion mechanism integrates aspect sentiment predictions with review credibility scores to generate reliable provider evaluation scores, enabling the selection of the final Top-5 trustworthy service providers.

The major contributions of this research are summarized as follows:

- Proposal of a Trust-Aware Sentiment Fusion mechanism that integrates aspect-level sentiment predictions with review credibility scores to reduce the influence of suspicious reviews during final provider ranking.

II. LITERATURE REVIEW

Recent advancements in Natural Language Processing (NLP) and deep learning have significantly improved sentiment analysis and recommendation systems by enabling automated extraction of meaningful information from textual customer reviews. Traditional machine learning techniques such as Naive Bayes, Support Vector Machines (SVM), and Logistic Regression primarily relied on handcrafted features, including Bag-of-Words (BoW) and TF-IDF representations, to classify customer opinions. Although these methods demonstrated acceptable performance, they were unable to effectively capture contextual relationships and semantic dependencies within review text, particularly for aspect-level sentiment analysis [6], [7].

To overcome these limitations, deep learning approaches have been widely adopted for sentiment classification. Bidirectional Long Short-Term Memory (Bi-LSTM) networks have shown superior performance by learning contextual information from both forward and backward directions of a sentence, enabling better understanding of customer opinions. Furthermore, the introduction of attention mechanisms has substantially improved sentiment classification by allowing models to focus on aspect-specific words that contribute most to prediction accuracy [2], [4]. More recently, transformer-based models such as BERT and Sentence-BERT have further enhanced contextual representation learning and semantic understanding in various sentiment analysis tasks [1], [5].

Aspect-Based Sentiment Analysis (ABSA) has emerged as an effective approach for analyzing customer opinions toward individual service quality dimensions rather than assigning a single sentiment label to an entire review. Deep learning approaches have been widely investigated for aspect-based sentiment analysis [11], [12], while multi-task learning enables related prediction tasks to share common representations and improve learning efficiency [10].

Another important research area is review credibility analysis and fake review detection. Online service platforms frequently suffer from review manipulation, opinion spam, and fraudulent feedback that negatively influence provider rankings and customer trust. Isolation Forest has become one of the most widely used unsupervised anomaly detection algorithms due to its ability to identify abnormal review patterns without requiring labelled datasets [8]. Previous studies have also investigated fake review detection using behavioural and linguistic features to distinguish genuine reviews from suspicious ones [9], [13]. Nevertheless, these approaches generally perform fake review detection as an independent task rather than integrating review credibility into sentiment-based provider evaluation.

However, existing approaches generally treat aspect-level sentiment analysis and review credibility detection as separate tasks, while limited attention has been given to integrating these signals for trust-aware service-provider ranking. To address this gap, this research proposes a Trust-Aware Review Intelligence Framework that combines Multi-Task Aspect-Based Sentiment Analysis, Isolation Forest-based Review Credibility Detection, and a Trust-Aware Sentiment Fusion mechanism. The proposed framework receives the Top-10 candidate providers from the previous recommendation stage, evaluates their customer reviews, and produces the final Top-5 trustworthy service providers.

III. METHODOLOGY

The proposed Trust-Aware Review Intelligence Framework serves as the final review analysis component of the ML-Driven Service Provider Matching Platform. The framework receives the Top-10 verified service providers generated by the Dynamic Credibility Verification component together with their associated customer reviews. Rather than relying solely on numerical ratings, the proposed framework

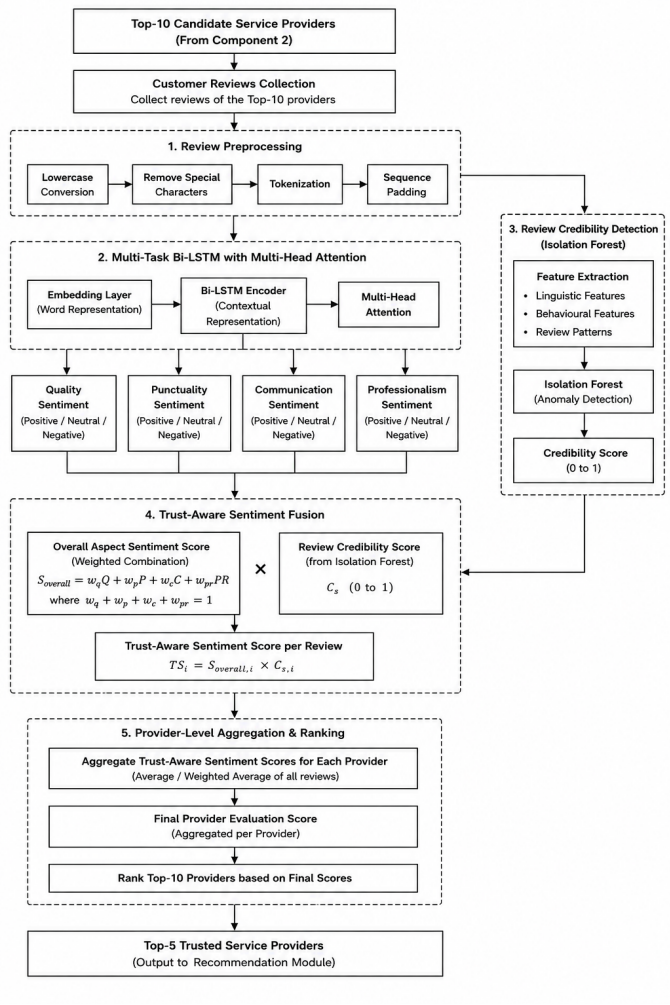


Fig. 1. Proposed Methodology Flowchart of the Trust-Aware Review Intelligence Framework

analyses customer opinions across multiple service quality aspects while simultaneously evaluating review credibility. The overall methodology consists of six sequential stages: review dataset preparation, review preprocessing, multi-task aspect-based sentiment analysis, review credibility detection, trust-aware sentiment fusion, and final provider ranking.

A. Review Dataset

A domain-specific dataset was developed to evaluate the proposed framework, as publicly available review datasets do not simultaneously provide aspect-level sentiment annotations and review credibility information for service provider recommendation systems. The dataset contains 25,000 customer reviews collected across fourteen service categories, including masonry, carpentry, plumbing, electrical services, technicians, painting, air-conditioning maintenance, cleaning services, vehicle repairs, CCTV installation, tile installation, ceiling works, well services, and cushion works.

B. Review Preprocessing

Reviews were lowercased, cleaned of noise, tokenized using the Keras Tokenizer, and converted into numerical sequences for model training.

Since customer reviews vary in length, all token sequences were padded to a fixed maximum sequence length using zero-padding. Finally, sentiment labels corresponding to each aspect were numerically encoded before model training. These preprocessing steps ensured standardized input representations while reducing noise and improving model performance.

C. Multi-Task Aspect-Based Sentiment Analysis

The proposed framework employs a Multi-Task Bidirectional Long Short-Term Memory (Bi-LSTM) network enhanced with a Multi-Head Attention mechanism to simultaneously predict customer sentiment across four service quality aspects. Unlike conventional single-task sentiment classifiers, the proposed model shares contextual representations through a common Bi-LSTM encoder while maintaining independent output layers for Quality, Punctuality, Communication, and Professionalism [4], [10], [11].

Initially, the preprocessed review sequence is converted into dense vector representations using an embedding layer before being forwarded to the Bi-LSTM encoder. The Bi-LSTM processes textual information in both forward and backward directions, enabling the model to capture contextual dependencies between words and improve semantic understanding of customer opinions [4]. Subsequently, the Multi-Head Attention mechanism assigns greater importance to aspect-relevant words, allowing the model to focus on opinion-bearing terms that significantly influence sentiment prediction [2].

The shared contextual representation is then forwarded to four independent softmax classification layers that simultaneously predict sentiment for each service quality aspect. This multi-task learning strategy improves feature sharing across related tasks while reducing computational complexity and improving model generalization compared with independently trained sentiment classifiers [10].

D. Review Credibility Detection

Although aspect-based sentiment analysis provides valuable insights into customer opinions, not all reviews can be considered equally reliable. Online service marketplaces frequently contain fake reviews, opinion spam, and manipulated feedback that may significantly influence provider rankings. Therefore, the proposed framework incorporates a Review Credibility Detection module to estimate the trustworthiness of each customer review before it contributes to provider evaluation.

The review credibility module employs the Isolation Forest algorithm, an unsupervised anomaly detection technique capable of identifying abnormal review patterns [8] without requiring manually labelled datasets. Several linguistic and behavioural features, including review length, word diversity, punctuation usage, capital letter ratio, repeated character patterns, reviewer activity, and rating behaviour, are extracted from each review and supplied to the Isolation Forest model

[9], [13]. Reviews exhibiting abnormal characteristics are assigned lower credibility scores, whereas genuine reviews receive higher credibility values. Consequently, suspicious reviews have a reduced influence during the final provider evaluation process [8], [9], [13].

E. Trust-Aware Sentiment Fusion

The proposed framework integrates aspect-level sentiment predictions with review credibility to reduce the influence of unreliable customer feedback during provider evaluation. For each review, the predicted sentiment scores for Quality (Q), Punctuality (P), Communication (C), and Professionalism (PR) are first combined using weighted aspect fusion:

$$S_{overall} = w_q Q + w_p P + w_c C + w_{pr} PR \tag{1}$$

where w_q, w_p, w_c, and w_{pr} represent the importance weights assigned to the four service-quality aspects, subject to:

$$w_q + w_p + w_c + w_{pr} = 1 \tag{2}$$

The overall sentiment score is then adjusted using the review credibility score (C_s) generated by the Isolation Forest-based credibility detection module:

$$TS = S_{overall} \times C_s \tag{3}$$

where TS denotes the Trust-Aware Sentiment Score of an individual review. This fusion mechanism ensures that reviews with positive aspect-level sentiment contribute strongly to provider evaluation only when they are also considered credible, while suspicious reviews receive a reduced influence on the final provider score.

F. Final Provider Ranking

Each service provider may receive multiple customer reviews. Therefore, the trust-aware sentiment scores generated from individual reviews are aggregated to calculate the final provider evaluation score.

$$ProviderScore = \frac{1}{N} \sum_{i=1}^{N} TS_i \tag{4}$$

where N represents the total number of customer reviews associated with a provider and TS_i denotes the Trust-Aware Sentiment Score of the ith review.

The Top-10 candidate providers received from Component 2 are ranked in descending order according to their calculated provider scores. The five highest-ranked providers are then returned as the final trustworthy service provider recommendations.

IV. SYSTEM DESIGN AND IMPLEMENTATION

A. System Architecture

The proposed framework operates as the final review intelligence module of the ML-Driven Service Provider Matching Platform. It receives the Top-10 candidate providers from Component 2 and analyses their customer reviews using Multi-Task Bi-LSTM with Multi-Head Attention and Isolation Forest-based credibility detection. The resulting sentiment and credibility scores are integrated through the Trust-Aware Sentiment Fusion mechanism to rank and select the final Top-5 trustworthy service providers.

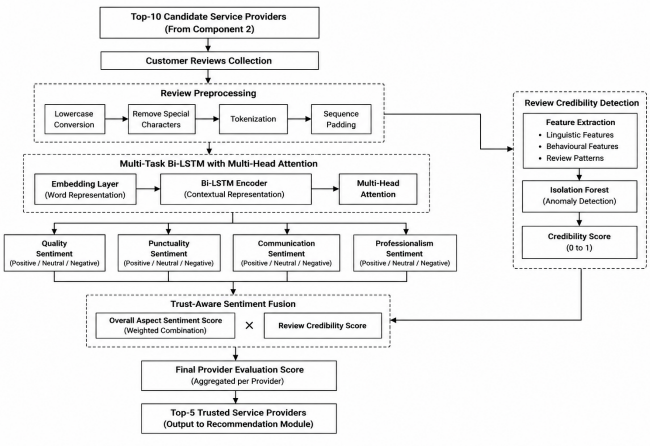


Fig. 2. System Overview Diagram

B. Framework Implementation

The framework was implemented in Python using TensorFlow/Keras for the Multi-Task Bi-LSTM model and Scikit-learn for Isolation Forest and evaluation metrics. Pandas and NumPy were used for data preprocessing and feature engineering. The trained models were integrated with the review intelligence pipeline to generate aspect-level sentiment predictions, review credibility scores, and final provider rankings.

C. Model Training Configuration

The Multi-Task Bi-LSTM model was trained using the Adam optimizer with sparse categorical cross-entropy as the loss function. A batch size of 64 and a maximum of 20 training epochs were used during model training. To improve model generalization and minimize overfitting, dropout regularization with a rate of 0.4, validation monitoring, and early stopping were incorporated throughout the training process.

The review dataset was divided into training, validation, and testing subsets using a 70:15:15 ratio. The validation dataset was utilized for hyperparameter tuning and model selection, while the independent testing dataset was reserved for final performance evaluation.

D. Experimental Environment

Table I summarizes the software environment and hyperparameter settings used to implement the proposed framework.

TABLE I
EXPERIMENTAL CONFIGURATION

Parameter	Value
Programming Language	Python 3.11
Development Environment	Google Colaboratory
Deep Learning Framework	TensorFlow / Keras
Machine Learning Library	Scikit-learn
Embedding Dimension	128
Bi-LSTM Units	128
Batch Size	64
Learning Rate	0.001
Epochs	20
Dropout Rate	0.40
Optimizer	Adam
Loss Function	Sparse Categorical Cross-Entropy

E. Review Credibility Detection Performance

The Isolation Forest-based credibility detection module demonstrated stable performance in identifying anomalous review patterns. On the independent test set containing 3,773 reviews, the model achieved an overall accuracy of 96%. For suspicious reviews, the model achieved a precision of 0.77, recall of 0.80, and F1-score of 0.79. The weighted F1-score across both real and suspicious review classes reached 0.96, indicating that the credibility detection module can effectively reduce the influence of anomalous review behaviour during provider evaluation.

The predicted credibility scores were subsequently integrated with aspect-level sentiment scores to generate trust-aware provider evaluation scores. Providers supported by higher credibility and trust-aware sentiment values achieved higher final ranking scores, demonstrating the influence of credibility-aware review aggregation on provider selection.

F. Dataset Analysis

Fig. 3 presents the statistical distribution of the annotated review dataset across the four service-quality aspects: Quality, Punctuality, Communication, and Professionalism. The balanced distribution of sentiment annotations across these aspects minimizes significant class imbalance and supports effective model training.

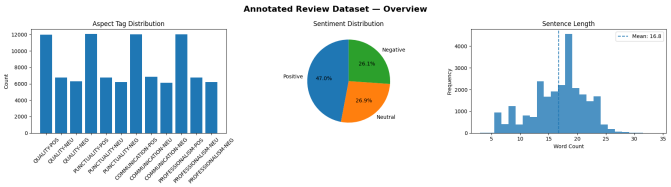


Fig. 3. Dataset Overview

G. Aspect-Level Sentiment Classification Performance

The proposed Multi-Task Bi-LSTM model successfully predicts customer sentiment for four service quality aspects simultaneously. Table II summarizes the experimental results obtained on the testing dataset.

The Quality aspect achieved the highest classification accuracy of 96.10% with a weighted F1-score of 96.11%, indicating that customer opinions regarding service quality are relatively easy to distinguish. The Punctuality aspect achieved an accuracy of 87.33%, while Communication and Professionalism obtained accuracies of 85.48% each. The proposed framework achieved an overall testing accuracy of 88.60%, demonstrating effective multi-task aspect prediction.

TABLE II
PERFORMANCE EVALUATION RESULTS

Aspect	Accuracy	Precision	Recall	F1-score
Quality	96.10	96.40	96.10	96.11
Punctuality	87.33	89.18	87.33	87.21
Communication	85.48	86.95	85.48	85.32
Professionalism	85.48	88.57	85.48	85.18
Average	88.60	-	-	88.46

H. Training Performance

Figs. 4 and 5 illustrate the training and validation accuracy and loss curves. Both metrics converge rapidly and remain stable during later epochs, while the limited divergence between training and validation results indicates stable optimization and no severe overfitting.

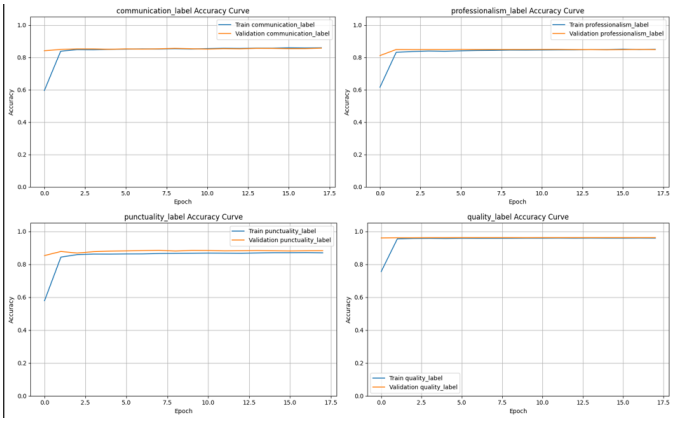


Fig. 4. Training and validation accuracy curves for the four aspect prediction tasks

I. Classification Analysis

The normalized confusion matrices shown in Fig. 6 indicate that the proposed model correctly classifies the majority of customer reviews across all sentiment categories. The Quality classifier achieved the highest prediction accuracy with minimal misclassification between Positive, Neutral, and Negative classes.

The Communication and Professionalism tasks exhibit comparatively higher confusion between Neutral and Negative sentiments, which is expected because customer opinions regarding these aspects often contain similar contextual expressions. Nevertheless, the overall classification performance remains consistent across all four tasks, confirming the effectiveness of the proposed multi-task architecture.

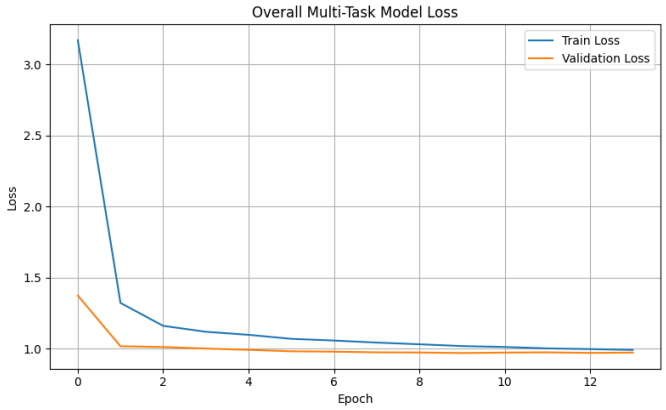


Fig. 5. Overall training and validation loss

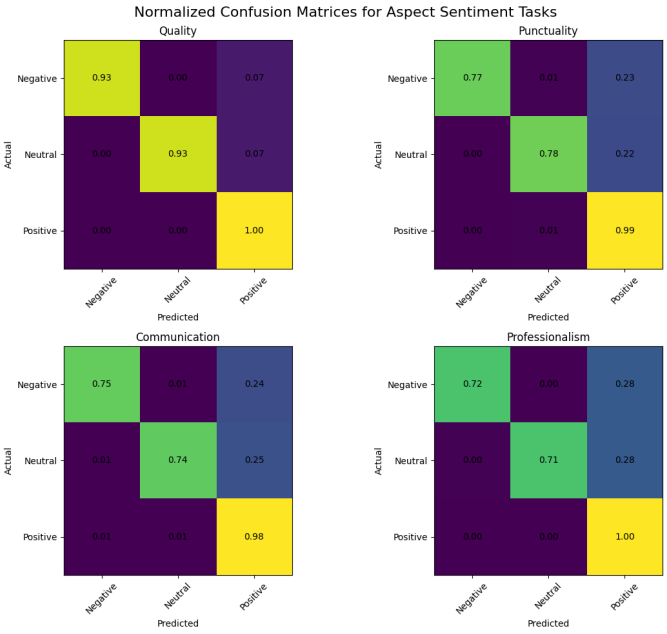


Fig. 6. Normalized confusion matrices for the four aspect sentiment prediction tasks

V. CONCLUSION

This paper proposed a Trust-Aware Review Intelligence Framework integrating Multi-Task Aspect-Based Sentiment Analysis, Isolation Forest-based Review Credibility Detection, and Trust-Aware Sentiment Fusion for reliable service provider evaluation. The Multi-Task Bi-LSTM with Multi-Head Attention simultaneously analyses customer sentiment across Quality, Punctuality, Communication, and Professionalism, while the credibility module reduces the influence of suspicious reviews during provider evaluation. Experimental evaluation on 25,000 customer reviews achieved an average testing accuracy of 88.60% across the four aspect sentiment classification tasks. The proposed framework combines these aspect-level predictions with review credibility information to support trust-aware ranking of candidate service providers. Future work will focus on multilingual review analysis, transformer-based architectures, and real-time deployment.

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